



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-044B ClinicQueue Protected Review

Open only after the matching Gate attempt/evidence is frozen.

Artifact	C3-044B-REVIEW
Status	2026 - LEARNER EDITION
Phase	C3-03 Orchestration & Reliable Workflows 13 lessons 34-46 focused hours
Prerequisite	C3-02 Gate PASS; C2B Python/quality/ownership foundations assumed, orchestration taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: simple mental picture -> tiny example -> guided Airflow/tool workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A green scheduler box alone never proves correct data.



Expected control shape

After chronological processing, current-state rows should be 6 and the current-state quantity x unit-value total should reconcile to 3450.00. Historical replay must not regress an entity whose later timestamp is already present.

Required orchestration properties

- Interval-derived partition; no wall-clock business date.
- Bounded readiness wait with timeout/efficient waiting.
- Bounded transient retry; deterministic bad data stays failed.
- Stable key/version rule; same-date rerun converges.
- Explicit backfill/reprocess range and active-run behavior.
- Case sink writes protected by named pool.
- Predeploy tests distinguish pure/static evidence from actual Airflow runtime.
- Recovery is minimal and followed by independent reconciliation.
- Runbook covers detect, diagnose, reprocess/rollback and validate.

Routing

PASS only when every required competency passes and zero critical failures remain. PARTIAL/FAIL -> narrow remediation + changed micro-retest. Fresh full Gate B/C opens only by explicit assignment; never automatically. Original failed attempts remain in history.